INCH-POUND
MIL-DTL-24643/20E
22 August 2002
SUPERSEDING
MIL-C-24643/20D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LS7SGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First -	Copper conductor, uncoated, (see table I for size)
Second -	Extruded silicone rubber insulation (see table I for diameter).
Third -	Glass braid.
Fourth -	Braid covering. Standard identification code applied by method 1
Fifth -	The seven conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly. Cabling sequence shall be consecutive starting with no. 1 from center outward.
Sixth -	A binder or combination binder and barrier.
Seventh -	Cross-linked polyolefin jacket. (see table I for thickness).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:00Z
Check the source to verify that this is the current version before use.

TABLE I. Details.

Military part no. M24643/20	Size	Conductor size AWG	Type's overall diameter (inches) LS7SGU min	Type's overall diameter (inches) LS7SGU max	Dia over insulation min (in.)	Dia over braid covering max (in.)	Jacket thickness min avg (in.)	Conductor resistance (ohms)
-01UN	3	16 (Class B)		0.545	0.096	0.130	0.040	4.3
-02UN	4	14 (Class B)		.595	.112	.143	.040	2.68

EXAMINATION AND TESTS:

Basic electrical: Requirements:

- Conductor resistance - ohms/1000 feet at 25°C, maximum ..... (see table I)
- Voltage withstand - volts, root mean square, minimum 3000
- Insulation resistance - megohms -1000 feet, minimum
  - Conductor to conductor 500
  - Conductor continuity ..... No failure

Group A:

- Visual and dimensional ..... No failure
- Watertightness - see MIL-DTL-24643 for limits of water leakage ..... No failure
- Abrasion resistance - minimum 250
- Crack resistance - ..... No damage

Group B:

- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket (When tested at 200°C) 50
- Gas flame - 1 hour ..... No failure
- Drip - 95 ± 1 °C ..... Zero
- Tear - pounds per inch thickness, minimum (ASTM D 470) 35
- Cold bending, cable - (maximum for mandrel diameter
  - 12 times maximum cable diameter ..... No damage

Physicals (unaged)

- Insulation (extruded)
  - Tensile strength - lb/in^2, minimum 750
  - Elongation - percent, minimum 125
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

Group C:

Physicals (aged)

- Jacket (cable)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:00Z
Check the source to verify that this is the current version before use.

Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (insulation) - cycles, minimum 25
Permanence of printing (jacket) - cycles, minimum 125
Cable sealant removability ..... No failure

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements:
Cold working (minus 20 ± 2°C)	No damage
Gas flame (3 hours)	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:00Z
Check the source to verify that this is the current version before use.

UNIT ORDERING LENGTHS:

All sizes 2000 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-018)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:00Z
Check the source to verify that this is the current version before use.